SURFACE AREA AND CUISENAIRE RODS SHEET 1

Problem: How many one-centimeter-by-one-centimeter unit squares does it take to cover all sides of each of the ten different Cuisenaire rods?

Solution plan: Take out the red and the white Cuisenaire rods. How many times must you "stamp" a face of the white rod on the faces of the red rod to cover that rod completely? _____ This result gives the total surface area of the red rod in square centimeters.

Repeat this "stamping" procedure for the remaining rods and record the results in the table below using the column "Total Surface Area."

Rod	Length	Total Surface Area	Lateral Surface Area
White	1 cm	6 cm^2	
Red	2 cm		
Light green			
Purple			
Yellow			
Dark green			
Black			
Brown			
Blue			
Orange			

Questions:

1. It is possible to determine the total surface area of a rod if you know the length of the rod. If a silver rod was 15 cm. long, what would its total surface area be? _____

   Write a formula that can be used to determine the total surface area T of a rod whose length is 1. T = _____ .

2. What would the surface area of a Cuisenaire rod be if you excluded the two ends? We call this area the lateral surface area of the rod. Complete the last column of the table above by determining the lateral surface area of each rod and entering the results in the appropriate positions in the table.

   Write a formula that can be used to determine the lateral surface area L of a rod whose length is *l*. L = _____ .

3. How are the two formulas alike?

4. How are the two formulas different?

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